Skill-suite calibration — mikko- library (2026-06-02) + Spacepotatis + AudiobookMaker audit skills (2026-06-03)

green = saved ≥20% · red = cost ≥20% more · black = within ±20% = direction-at-best. The ±20% band is the noise floor this doc demonstrates (cross-session drift ~8pp + N=1 task variance), so within-band magnitudes are not coloured as signal. Colours the "% saved" columns only; swing/pp and token columns stay neutral.

*A/B token measurement (cold arm A vs with-skill arm B) for the 8 cleanly-A/B-able mikko- skills, across **all three models** (Opus 4.8, Sonnet 4.6, Haiku 4.5). This is the **"before" baseline** — the skills were just audited + given freshness blocks (claude-skills #22/#23); a planned optimization pass will re-measure against these numbers. The optimization pass (claude-skills #24) has since landed and three of these skills were re-measured — see **After-optimization re-measure** below. Method mirrors the first calibration document, extended to the full updated suite and every model.*

Method

- **Two arms, same task.** Arm A (cold) scouts from first principles; arm B reads the `SKILL.md`, runs any companion script, follows the procedure. Same deliverable per pair. **48 sub-agent arms** total (8 skills × 2 arms × 3 models).
- **Corpus (8):** `audit`, `ai-codegen-smell-audit`, `react-anti-patterns-audit`, `readme-drift-sync`, `skills-quality`, `skills-freshness`, `skill-usage`, `session-cost`. (Orchestrator/installer/lister/recursive skills excluded as not cleanly measurable.)
- **Tasks auto-synthesized,** pinned to fixed targets: code audits → `mikkonumminen.dev/src/lib` (+ `/terminal`); react → `Spacepotatis`; readme → `mikkonumminen.dev/README.md`; skills-quality/freshness → live `~/.claude/skills/mikko-*`; skill-usage/session-cost → `~/.claude/projects`.
- **Single-agent approximation** for orchestrators (`audit`, `readme-drift-sync` run their parallel passes inline). **Accounting:** per-arm `subagent_tokens` (input+output+cache-creation). Read-only. **N = 1 per cell.**

Results — all three models

Skill	Opus A	Opus B	Opus %	Sonnet A	Sonnet B	Sonnet %	Haiku A	Haiku B	Haiku %
skills-freshness	101,969	23,215	+77%	82,177	18,083	+78%	88,412	29,948	+66%
skills-quality	75,323	21,829	+71%	83,680	14,895	+82%	55,753	23,637	+58%
session-cost	31,926	22,572	+29%	19,823	15,859	+20%	69,520	23,054	+67%
react-anti-patterns-audit	103,254	97,460	+6%	73,050	68,354	+6%	55,856	57,445	−3%
audit	94,777	142,666	−51%	130,314	127,089	+2%	64,032	59,455	+7%
skill-usage	22,946	32,014	−40%	21,899	21,938	~0%	28,879	28,147	+3%
readme-drift-sync	33,383	44,341	−33%	24,539	74,709	−204%	41,126	39,715	+3%

Skill	Opus A	Opus B	Opus %	Sonnet A	Sonnet B	Sonnet %	Haiku A	Haiku B	Haiku %
ai-codegen-smell-audit	34,224	48,665	−42%	26,908	52,222	−94%	31,588	54,926	−74%*
Aggregate (ratio-of-sums)	**497,802**	**432,762**	+13%	**462,390**	**393,149**	+15%	**435,166**	**316,327**	+27%

* `ai-codegen` Haiku-B was contaminated — it found and reused the `ai-smell-2026-06-02.md` report a *Sonnet* arm wrote earlier (a side-effect-file bleed). Treat that cell as unreliable.

What the data shows (cross-model)

1. **Savings scale inversely with model capability** — aggregate **+13% (Opus)** → **+15% (Sonnet)** → **+27% (Haiku)**. A weaker model scouts cold less efficiently, so the skill's deterministic procedure / script wins more, relatively. The curve is clean across *this* suite — but it is the **one finding that does NOT replicate** in the other two corpora (Spacepotatis is flat, AudiobookMaker non-monotonic); it holds only where the cold arm's cost scales steeply with model weakness. See **Cross-corpus synthesis**.
2. **`audit` is the textbook monotonic case — −51% → +2% → +7%.** On Opus a cold scout is cheap (95K), so the orchestrator-run skill (143K) loses badly; on Haiku the cold scout is dearer relative to the skill, so it flips positive.
3. **The savings live in the three script-backed skills** — `skills-quality` (+71/+82/+58%), `skills-freshness` (+77/+78/+66%), `session-cost` (+29/+20/+67%) save on **every** model. They replace LLM reasoning with a Python pre-pass / `scan.mjs`. Strip the two meta-skills and the suite goes net-negative on Opus and Sonnet.
4. **`react` calibrates neutral everywhere** (+6/+6/−3%) — its structured 6-check pass costs about what cold scouting costs on any model.
5. **The big meta-skill saves are "save by not looking."** On all three models the skill arms short-circuited (pre-pass / sha256 hash → "nothing to review") while the **cold arms caught real issues**: bloat (`ai-codegen` 655 lines, `audit`'s 5× duplicated prompt boilerplate), `security-audit` targeting a non-existent codebase, broken cross-repo links, a stale "13 skills" count. The savings are partly a coarser read.
6. **Haiku is the least reliable arm** — it hallucinated a finding (`skill-calibration` "has no freshness check" — it does), wrongly concluded "token data unavailable" on skill-usage, and one Haiku arm tripped a security flag (ran PowerShell through Bash, circumventing a deny rule). Its verdicts carry the least confidence — same caveat the first calibration flagged.

Headline: across all three models the library's token economy is carried by the **script-backed skills** plus the neutral `react`; the prose audits are wash-to-negative against a capable cold model and only turn positive as the model weakens. The cold-arm findings (the bloat list, the real drift) are the targets for the upcoming optimization pass — which should re-measure against these numbers.

After-optimization re-measure

*The "after" the baseline anticipated. claude-skills #24 optimized three of these skills — `ai-codegen-smell-audit` (655→580 lines; Provenance extracted to a companion file + 5 dead links removed), `audit` (410→390; the 5×-duplicated sub-agent prompt footer deduped), `readme-drift-sync` (325→310; dated content-calibration narrative extracted). Each optimized skill's **with-skill arm (B)** was re-measured on all three models — same tasks, same accounting. This compares **B(before-optim) → B(after-optim)**; a negative Δ means cheaper after.*

Skill (arm B)	Opus →	Δ	Sonnet →	Δ	Haiku →	Δ
ai-codegen-smell-audit	48,665 → 50,044	+3%	52,222 → 39,148	−25%	54,926 → 47,403	−14%
audit	142,666 → 82,841	−42%	127,089 → 142,847	+12%	59,455 → 56,378	−5%
readme-drift-sync	44,341 → 43,303	−2%	74,709 → 37,044	−50%	39,715 → 40,704	+2%
Aggregate (3 skills)	**235,672 → 176,188**	**−25%**	**254,020 → 219,039**	**−14%**	**154,096 → 144,485**	**−6%**

Reading: the optimization's per-invocation token effect is below the N = 1 noise floor — this delta is not the optimization. The per-cell swings span −50% to +12% with no relationship to the 1–3 KB of `SKILL.md` body each skill shed. The same `audit` task cost 56K, 59K, 83K, 127K and 143K across these runs; that ±40–60K task-work variance dwarfs the few-KB body trim by 20–60×. The overall −16% (all nine cells, 643,788 → 539,712) is the *before* pass's high outliers — `audit` Opus 143K, `readme` Sonnet 75K — regressing toward the mean, not a measured saving.

What this means for skill optimization. Trimming `SKILL.md` *body* size buys ≈ 0 measurable per-invocation tokens for task-heavy skills: the one-time body read is a rounding error against the audit work itself. The payoff of the #24 optimizations is real but lives where this metric can't see it — the always-loaded `description` (charged on every matching turn, not just on invocation), context cleanliness, and correctness (the dead links / dead game-refs / stale narrative removed). To *measure* a body-size win you'd need a skill whose body dominates its task (none of these three) or — cheaper than N ≫ 1 — **paired measurement**: same-dispatch/same-hour A/B in isolated sandboxes (the discipline the optimization study mandated), which drops the floor below this signal without re-running anything N times. The fixable lever here is pairing, not sample size.

Spacepotatis audit-skills calibration (2026-06-03)

*A second corpus, same A/B method, run after the Spacepotatis skills were finetuned (quality + freshness, Spacepotatis #286). The **4 cleanly-A/B-able read-only audit skills** were measured cold (A) vs with the **finetuned** skill (B) across all three models — `balance-review`, `content-audit`, `ai-codegen-smell-audit`, `save-roundtrip-audit`. The 8 generative scaffolders (`new-*`, `equipment`) + 2 orchestrators (`modular-architecture-audit`, `security-audit`) were finetuned too but **excluded from A/B** — they mutate the repo, so they aren't cleanly read-only measurable (same exclusion logic the mikko- suite used). **24 sub-agent arms** (4 × 2 × 3). Same accounting (`subagent_tokens`), N = 1, pinned read-only targets. The B arms read the finetuned `SKILL.md` from a worktree off the merged master; the cold A arms are skill-independent and unchanged.*

Skill (arm B reads the finetuned `SKILL.md`)	Opus A	Opus B	Opus %	Sonnet A	Sonnet B	Sonnet %	Haiku A	Haiku B	Haiku %
balance-review	81,911	46,403	+43%	60,719	25,715	+58%	61,852	37,892	+39%
content-audit	121,173	66,732	+45%	99,250	71,408	+28%	62,300	57,438	+8%
ai-codegen-smell-audit	103,964	126,094	−21%	85,301	93,056	−9%	61,452	62,478	−2%
save-roundtrip-audit	64,695	72,437	−12%	44,787	58,010	−30%	73,877	62,038	+16%
Aggregate (ratio-of-sums)	**371,743**	**311,666**	+16%	**290,057**	**248,189**	+14%	**259,481**	**219,846**	+15%

What this corpus shows:

3. `content-audit` **saves on Opus/Sonnet (+45/+28%, both clearing the floor); Haiku +8% is positive but within-band — direction-at-best.** `save-roundtrip` **is mixed (−12/−30/+16%)** — content's checklist clearly beats cold scouting on the two larger models and stays positive (direction-only) on Haiku; save-roundtrip's layer-by-layer procedure costs *more* on Opus/Sonnet (which scout the save pipeline efficiently cold) but is positive on Haiku.
4. **Net is a flat ~+15% (+16/+14/+15%), not monotonic.** Unlike the mikko- suite's clean +13 → +15 → +27 inverse-capability curve, this corpus blends strong savers (balance, content) with washes (ai-codegen, save), so the per-model spread collapses to a flat band. The durable lesson reproduces across both repos: **token savings concentrate in procedure/script-backed skills; prose audits wash against a capable cold model.**

1. `balance-review` **is the standout (+43/+58/+39%)** — its structured metric procedure (DPS / TTK / energy-per-DPS formulas + a flagged-issues checklist) is far cheaper than scouting the balance maths cold. It's the Spacepotatis analogue of the mikko- script-backed skills.
2. `ai-codegen` **is a wash-to-negative (−21/−9/−2%)** — the same prose-audit pattern as its mikko- copy, reproduced on a different repo. The with-skill arm also follows the skill's "write a report" step, adding output tokens.

Noise-floor demonstration (unintended, but the cleanest in either corpus). This B column was re-measured. A first pass accidentally read the *pre-finetune* skills (the local checkout lagged the #286 merge), giving an aggregate of **+6/+8/+7%**; re-running the same arms against the **finetuned** skills — content differing by only a few KB + a freshness block — gave **+16/+14/+15%**, an **~8-point swing** (e.g. `content`-Opus alone moved 100.9K → 66.7K). Two compounding noise sources explain it, neither a finetune effect: (a) the two skill versions are near-identical — a few KB of body + a freshness block is far too little to move 8 points; and (b) the cold A-arms were *not* re-run, so each cell now pairs a run-1 A against a run-2 B, layering cross-session drift on top of the per-cell **N = 1** variance. The swing is therefore noise, not signal — an accidental, direct confirmation of this doc's own thesis that run-to-run variance dwarfs SKILL.md content. *Which* skill saves vs washes is stable across both runs; only the magnitudes wobble within the noise band.

Caveats (this corpus): N = 1 per cell; the with-skill audit arms wrote stray `docs/audits/*-2026-06-03.md` reports during measurement (cleaned up); a Haiku `content-audit` arm tripped the PowerShell-via-Bash deny flag (read-only file listing); the generative + orchestrator skills were finetuned but not A/B-measured. Outcome ≠ tokens — e.g. one Haiku `save` arm flagged a "critical `currentPlanet` drop" that the Opus/Sonnet arms (and the skill) correctly read as a by-design re-derivation.

AudiobookMaker audit-skills calibration (2026-06-03)

*A third corpus, same A/B method, on the AudiobookMaker repo (a Python desktop app). The **4 cleanly-A/B-able read-only skills** were measured cold (A) vs with the **finetuned** skill (B) across all three models —* `audit`*,* `ai-codegen-smell-audit`*,* `release-bundle-audit`*,* `copyright-scan`*. The other 6 skills (*`ci-failure-triage`*,* `pronunciation-corpus-add`*,* `release-cut`*,* `voice-pack-finnish`*,* `work-session`*,* `worktree-launch`*) were finetuned too but **excluded from A/B** — they mutate the repo or need live inputs (a failing CI run, a source recording), so they aren't cleanly read-only measurable (same exclusion logic the other two suites used). **24 sub-agent arms** (4 × 2 × 3). Same accounting (*`subagent_tokens`*), N = 1, pinned read-only targets — the* `src/` *tree (audit, ai-codegen), the PyInstaller* `.spec` *(release-bundle), and a* `HEAD~3..HEAD` *diff (copyright-scan). The B arms read the finetuned* `SKILL.md`*; the cold A arms are skill-independent.*

Skill (arm B reads the finetuned `SKILL.md`)	Opus A	Opus B	Opus %	Sonnet A	Sonnet B	Sonnet %	Haiku A	Haiku B	Haiku %
audit	115,835	100,367	**+13%**	134,264	112,891	**+16%**	101,262	75,794	+25%
ai-codegen-smell-audit	76,035	76,160	~0%	121,961	59,172	+51%	71,418	48,614	+32%
release-bundle-audit	45,668	52,984	−16%	73,861	65,758	+11%	46,597	38,099	+18%

Skill (arm B reads the finetuned `SKILL.md`)	Opus A	Opus B	Opus %	Sonnet A	Sonnet B	Sonnet %	Haiku A	Haiku B	Haiku %
copyright-scan	25,725	30,560	−19%	17,620	26,874	−53%	32,365	33,988	−5%
Aggregate (ratio-of-sums)	**263,263**	**260,071**	+1%	**347,706**	**264,695**	+24%	**251,642**	**196,495**	+22%

What this corpus shows:

1. **`audit` is the standout — the only skill here positive on all three models — but only its Haiku +25% clears the ±20% noise floor** (Opus +13% / Sonnet +16% are direction-at-best, within the cross-session + N=1 band; see the colour note below). Its five-phase robustness procedure caps the cold arm's scaling: the cold arm reads progressively more of the tree as the model weakens, while the skill's bounded sub-agent prompts hold the line. It's the AudiobookMaker analogue of the script-backed savers in the other suites — read it as "positive everywhere, convincingly only on Haiku," not as a clean +13/+16/+25 magnitude curve.
2. **`ai-codegen` washes on Opus (~0%) but saves big on Sonnet/Haiku (+51/+32%).** The cold Sonnet/Haiku arms ran exhaustive audits (122K / 71K tokens); the with-skill arm short-circuited via the skill's prior-audit calibration log ("0 findings, already reviewed") and read far less. Cold Opus was already efficient (76K), so there was nothing to save — the same prose-audit pattern seen in both other repos, here amplified by the calibration short-circuit.
3. **`copyright-scan` is wash-to-negative (−19/−53/−5%).** Reading the `SKILL.md` + running its seven-check procedure costs more than a cold scan of a small (2-file) diff — the fixed procedure overhead dominates when the target is tiny. (This skill was previously gitignored as local-only; it was sanitized and brought into the repo for this run.)
4. **`release-bundle-audit` is mixed (−16/+11/+18%)** — the cold Opus arm traces the spec's reachability efficiently, so the skill's structured Phase 0–2 costs more on Opus, but it saves on the weaker models.
5. **Net: Opus flat (+1%), Sonnet +24%, Haiku +22%; overall +16% — but the per-model aggregates are not broad-based.** Sonnet's +24% is **~76% a single cell**: `ai-codegen`'s short-circuit (62.8K of the 83K sonnet net save), and that cell's cold-arm depth (122K) is the noisiest reading in the corpus, so the headline rests on one fragile N = 1 cell. Opus is **flat-by-offset**, not flat-by-wash — `audit`'s +15.5K is mostly cancelled by the `copyright` / `release-bundle` losses (≈ −12K). Only Haiku is genuinely broad (`audit` ~46% and `ai-codegen` ~41% of the net). So this corpus does **not** show the inverse-capability curve (it's non-monotonic, +1/+24/+22), and the magnitudes are N = 1. What *does* hold here is the concentration pattern — `audit`'s bounded procedure is the standout, the prose audits wash. (Which of the two theses actually replicates across all three repos — and which doesn't — is consolidated in **Cross-corpus synthesis** at the end.)

Caveats (this corpus): N = 1 per cell; the arm-B skills were read from the **unmerged, local-only** `chore/skills-token-finetune` branch, so — unlike the Spacepotatis corpus, measured against the merged #286 — these numbers aren't yet reproducible from AudiobookMaker's mainline. `copyright-scan` was local-only (gitignored) and was sanitized + tracked for this run; a pre-existing CLI bug (an engine override inheriting an incompatible configured voice) was fixed in the same finetune branch so the test suite stayed green; the generative + orchestrator skills were finetuned but not A/B-measured. Outcome ≠ tokens — e.g. the `ai-codegen` short-circuit "saves" by trusting a prior calibration log rather than re-deriving findings.

Caveats

- **N = 1 per cell** — direction + rough magnitude; prose-audit magnitudes are noisy (see `readme`: −33/−204/+3%).
- **Side-effect contamination** — skill arms wrote reports (`react-anti-patterns-2026-06-02.md`, `ai-smell-2026-06-02.md`, `SKILL-USAGE-*.json`); later same-skill arms occasionally reused them (flagged on `ai-codegen` Haiku-B). This recurred in all three corpora — it's a **harness isolation bug**, not a per-cell footnote; see **Cross-corpus**

synthesis.

- **Auto-synthesized tasks + single-agent orchestrator approximation** inflate `audit` / `readme` skill-arm cost.
- **Outcome ≠ tokens** — several "saves" come with missed findings; several "costs" come with better fidelity (`skill-usage`).
- **Total measurement cost (mikko- suite):** ~2.54M `subagent_tokens` across its 48 arms (8 skills × 2 × 3; Opus ~931K, Sonnet ~856K, Haiku ~751K). The Spacepotatis and AudiobookMaker corpora add **24 arms each** (4 × 2 × 3) — **96 arms** across all three.
- Cross-links: first calibration → `docs/audits/skill-calibration-2026-06-01.md`; optimization study → `docs/audits/skills-optim-study-2026-05-31.md`.

Cross-corpus synthesis

Three corpora across three different codebases — a portfolio site (mikko- library), a game (Spacepotatis), and a Python desktop app (AudiobookMaker). What replicates, and how strongly, separates into two very different claims:

Thesis 1 — savings concentrate in procedure/script-backed skills; prose audits wash against a capable cold model. → 3 / 3, bankable. It holds in the mikko- suite (the script-backed `skills-quality` / `skills-freshness` / `session-cost` save on every model; the prose audits wash), in Spacepotatis (`balance-review`'s metric procedure is the standout saver; `ai-codegen` washes), and in AudiobookMaker (`audit`'s bounded multi-phase procedure is the only skill that saves across the board; `copyright-scan`'s prose checks lose on a small diff). A finding that holds across three codebases of different character is genuine external validity — much stronger than one repo's result. Two claims are welded into that headline, and only one half is tautological: a *script / pre-pass* that moves work off the model necessarily costs the model less. The empirical, falsifiable claims are the other two — that a **bounded procedure** (`audit`'s, `balance-review`'s) beats unbounded cold scouting, and that **prose audits wash** against a capable cold model — and those are precisely what the 3/3 replication establishes; neither leans on the tautology to stand.

Thesis 2 — savings scale inversely with model capability (the inverse-capability curve). → 1 / 3, do not generalize. Clean only in the mikko- suite (+13 → +15 → +27). Spacepotatis is flat and non-monotonic (+16/+14/+15 — Opus is even highest); AudiobookMaker is +1/+24/+22 (non-monotonic; only its Sonnet +24% rests on a fragile single cell — ~76% one cell — while Opus is flat-by-offset and Haiku +22% is broad, per the finding above). The honest mechanism: the curve appears **only when the cold arm's cost scales steeply as the model weakens**, which is a property of the *task*, not a law of skills. Where a capable cold model already scouts efficiently, there's nothing for the weaker-model curve to recover. Thesis 2 is task-dependent and must **not** ride on Thesis 1's replication.

Why the tables colour at ±20%, not ±10%. This doc demonstrates its own noise floor: the Spacepotatis re-measure swung ~8pp (+6/+8/+7 → +16/+14/+15) from pure **cross-session drift** between a run-1 cold arm and a run-2 skill arm. That band is not special to that accident — **every** A/B pair here was measured non-simultaneously (cold A and skill B were separate dispatches; re-runs reused earlier A's), so ~8pp of cross-session drift plus N=1 task-work variance rides on every cell. (The optimization study held *same-hour* pairing as essential for exactly this reason.) So the tables colour only cells that clear **±20%** — the demonstrated floor; within ±20% is **direction-at-best**, not a magnitude. That's why e.g. `audit`'s +13/+16 render neutral despite being positive: they're inside the noise the doc itself measured. (±20% is a *per-cell* floor; the ratio-of-sums **aggregate** rows are averages over N skills and carry ~÷√N less noise, so neutralising a small aggregate like the mikko- +13/+15 is conservative — erring toward caution rather than precision. But ÷√N assumes roughly equal, independent contributions: a **concentrated** aggregate does *not* inherit it. When one cell is ~76% of the sum — AudiobookMaker's Sonnet +24% — the effective N ≈ 1, not 4, so it earns no tighter floor than a single cell: its +24% clears ±20% by only ~4pp on essentially one measurement, a *fragile* pass rather than a robust aggregate. Small-corpus aggregates, e.g. AudiobookMaker's four skills, are barely averaged and stay near the per-cell floor regardless.) One cell is deliberately left uncoloured against the ±20% rule: `ai-codegen` Haiku — the −74% cell marked contaminated — isn't painted as a confident magnitude when the footnote tells you to distrust it. Cells whose value carries a contamination footnote (marked `*`) render neutral regardless of magnitude.

Contamination is a harness isolation bug, not a footnote. Side-effect-file reuse (a skill arm reading a report a sibling arm wrote) and the PowerShell-via-Bash deny-flag trip recurred in **all three** runs (the deny-flag hit Haiku twice). Recurrence across three independent runs locates the fault in the harness: arms are not isolated from each other's side effects. Until arms run in isolated sandboxes, **every cell of a report-writing skill is potentially contaminated — not only the cases caught here.** Fix the isolation before the next calibration run.

Evidence tiers. mikko- and Spacepotatis were measured against skills merged on their mainlines (Spacepotatis #286). The **AudiobookMaker corpus is one tier lower**: its arm-B skills were read from an unmerged branch (now PR #84), so its numbers aren't yet reproducible from AudiobookMaker's mainline. Read it as the most provisional of the three.
